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# SAGCO-CLI v0.1.0
**Field Engineering Productivity CLI**
Strategickhaos DAO LLC | EIN 39-2900295 | Dom Garza
```

What This Does

Computes Expected vs Actual vs Variance for industrial insulation rope access projects. Takes a YAML config with circuit data, outputs CPI, remaining hours, savings, and per-circuit audit.

Build

Requires Rust 1.80+

```
```bash
cargo build --release
```
```

Commands

```
```bash
Summary metrics
./target/release/sagco-cli compute --config sagco.master.yaml

Per-circuit audit with ERU analysis
./target/release/sagco-cli audit --config sagco.master.yaml

Validate config file
./target/release/sagco-cli validate --config sagco.master.yaml
```
```

Expected Output – compute

```

SAGCO COMPUTE   C-1602 Package   PO 4406147216		
Est Hours	:	4979.0
Act Hours	:	3583.3
Remaining	:	1395.7
CPI	:	1.390
Variance	:	-28.0%
Savings @\$77:		\$107,468.90
STATUS	:	AHEAD

```

Expected Output – audit

```

...

```

| CUI | CIRCUIT | EST | ACT | REMAIN | VAR% | \$AT RISK |
|---------------|---------------|--------|--------|--------|---------|--------------|
| VERDICT | | | | | | |
| 1 | F-1763-PR-1A | 290.0 | 427.5 | -137.5 | -47.4% | \$10,588 |
| OVER | | | | | | |
| 2 | C-1602-O-1B | 1080.0 | 1118.4 | -38.4 | -3.6% | \$2,957 |
| OVER | | | | | | |
| 3 | IS-HDRFE-1A | 491.0 | 1020.3 | -529.3 | -107.8% | \$40,765 |
| CRITICAL OVER | | | | | | |
| 4 | E-1706-NA-1A | 138.0 | 216.9 | -78.9 | -57.2% | \$6,075 |
| OVER | | | | | | |
| 5 | PO-HDRFE-1B | 204.0 | 173.3 | 30.7 | 15.0% | BANKED \$2k |
| UNDER | | | | | | |
| 6 | HBF-HDRULT-1C | 1021.0 | 208.3 | 812.7 | 79.6% | BANKED \$63k |
| BANKED | | | | | | |
| 8 | J-2011-FO-1B | 951.0 | 164.6 | 786.4 | 82.7% | BANKED \$61k |
| BANKED | | | | | | |
| ... | | | | | | |

```

Overrun hours : 784.1 ($60,375.70)
Banked hours  : 2179.8 ($167,844.60)
Net position  : $107,468.90
...

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## Config Format

```

```

```yaml
project: "C-1602 Package"
po: "4406147216"
blend_rate: 77.0
difficulty_multiplier: 1.69
circuits:
 - cui_number: "1"
 circuit_id: "F-1763-PR-1A"
 est_hours: 290.0
 act_hours: 427.5
 lnft: 75.0
 sqft: 649.2
 status: "COMPLETE"
...

```

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Project Data - PO 4406147216

```

```

| Field | Value |
|---|---|
| Project | C-1602 Package |
| PO | 4406147216 |
| Est Hours | 4,979.0 |
| Act Hours | 3,583.3 |
| Remaining | 1,395.7 |

```

```
| Blend Rate | $77/hr |
| Est $ | $483,228 |
| Act $ | $333,532 |
| Variance $ | +$149,696 |
| CPI | 1.390 (AHEAD) |
| Circuits | 12 |
```

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## Install on Termux (Z Fold)

```
```bash
pkg install rust
tar -xzf sagco-cli.tar.gz
cd sagco-cli

# IMPORTANT - if sagco-cli already exists as a git repo:
cp src/*.rs ../sagco-cli-new/src/
# or just overwrite the source files manually
```

```
cargo build --release
./target/release/sagco-cli compute
```
```

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## Source Tree

```
```
sagco-cli/
├── Cargo.toml
├── README.md
├── sagco.master.yaml
├── src/
│   ├── main.rs
│   ├── config.rs
│   ├── models.rs
│   ├── commands/
│   │   ├── mod.rs
│   │   ├── compute.rs
│   │   ├── audit.rs
│   │   └── validate.rs
│   └── engines/
│       ├── mod.rs
│       ├── productivity.rs
│       └── eru.rs
```
```

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## The Fix for "Hello, world!" Output

If you extracted the tar into an existing sagco-cli git repo,  
the old src/main.rs overwrote the new one. Run this on Termux:

```
```bash
cd ~/downloads/sagco-cli
```

```

cat > src/main.rs << 'RUST'
use clap::Parser;
use anyhow::Result;
mod config; mod models; mod engines; mod commands;
#[derive(Parser)]
#[command(about = "SAGCO Field Engineering CLI")]
struct Cli { #[command(subcommand)] command: Commands }
#[derive(clap::Subcommand)]
enum Commands {
    Compute(commands::compute::ComputeArgs),
    Audit(commands::audit::AuditArgs),
    Validate(commands::validate::ValidateArgs),
}
fn main() -> Result<()> {
    let cli = Cli::parse();
    match cli.command {
        Commands::Compute(a) => commands::compute::execute(a),
        Commands::Audit(a)   => commands::audit::execute(a),
        Commands::Validate(a) => commands::validate::execute(a),
    }
}
RUST

```

```

cargo build --release
./target/release/sagco-cli compute
\`

```

Inventor

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